

PAT1:-

Single File Programming Question
Problem Statement

Nilu needs to calculate the income tax based on a specific tax rate system.

Write a program to determine the tax bracket and calculate the tax amount based on Nilu's income using the tax rate system.

High Income (Income > 200,000):

If the income is greater than 500,000:

The tax rate is 35%.

Otherwise:

The tax rate is 30%.

Moderate Income (Income greater than or equal to 50,000):

If the income is greater than 100,000:

The tax rate is 20%.

Otherwise:

The tax rate is 15%.

Low Income (Income less than 50,000):

The tax rate is 10%.

Input format :

The input consists of a double value N, representing Nilu's income.

Output format :

The first line of output prints "The tax bracket is: X%", where X is the tax rate percentage.

The second line prints "The calculated tax is: \$Y", where Y is the calculated tax amount.

Refer to the sample output for formatting specifications.

Code constraints :

$3000.0 \leq N \leq 6000000.0$

Sample test cases :

Input 1 :

30000.0

Output 1 :

The tax bracket is: 10%

The calculated tax is: \$3000.0

Input 2 :

60000.0

Output 2 :

The tax bracket is: 15%

The calculated tax is: \$9000.0

Question No: 2

Single File Programming Question
Problem Statement

Liam is organizing a community event and allocates 30% of his total funds for decorations and 20% for refreshments. Write a program to calculate the remaining amount after these expenses. Input the total funds and output the remaining amount formatted to two decimal places.

Input format :

The input consists of a double value B, representing the total funds for the event.

Output format :

The output displays a double value, representing the remaining amount rounded to two decimal places.

Refer to the sample output for formatting specifications.

Code constraints :

$1000.00 \leq B \leq 10000.00$

Sample test cases :

Input 1 :

1000.00

Output 1 :

500.00

Input 2 :

4504.20

Output 2 :

2252.10

PAT2:-

Question No: 1

Single File Programming Question

Problem Statement

Aaradhya has an array of integers representing her collection. She wants to find the largest and smallest elements from her collection. Help her by developing a program that can find and display such elements.

Input format :

The first line of input consists of an integer n, representing the number of elements in the collection.

The second line consists of n space-separated integers, representing the elements in Aaradhya's collection.

Output format :

The first line of output prints "The largest element is: " followed by the largest element in the collection.

The second line prints "The smallest element is: " followed by the smallest element in the collection.

Refer to the sample output for formatting specifications.

Code constraints :

$1 \leq n \leq 10$

$1 \leq \text{elements in collection} \leq 1000$

Sample test cases :

Input 1 :

5

89 68 57 54 25

Output 1 :

The largest element is: 89

The smallest element is: 25

Input 2 :

10

78 96 75 63 62 45 52 21 23 28

Output 2 :

The largest element is: 96

The smallest element is: 21

Question No: 2

Single File Programming Question
Problem Statement

A mobile app is being developed to analyze user phone numbers and identify if they consist only of prime digits. In this context, the prime digits are defined as 2, 3, 5, and 7. The app will process a list of phone numbers to determine if each number contains only these prime digits.

Write a program that takes a list of phone numbers and determines if each phone number contains only the prime digits. For each phone number, the program should output 1 if all digits are prime, and 0 otherwise.

Input format :

The first line contains an integer T, the number of phone numbers to process.

The next T lines each contain a phone number represented as an integer N.

Output format :

For each phone number, output 1 if all digits are prime, otherwise output 0.

Refer to the sample output for formatting specifications.

Code constraints :

$1 \leq T \leq 10$

$0 \leq N \leq 3000$

Sample test cases :

Input 1 :

2

233

240

Output 1 :

1

0

Input 2 :

3

555

973

111

Output 2 :

1

0

0

PAT3:-

Question No: 1

Single File Programming Question
Problem Statement

You are tasked with calculating the total production cost of cakes ordered for a bakery. Each cake has a name, a unit cost, and a quantity ordered. After calculating the total cost of all cakes, apply the following discounts based on the total cost before discount:

20% discount if the total cost is more than 100.00 dollars.

10% discount if the total cost is more than 50.00 dollars but less than or equal to 100.00 dollars.

Requirements

Cake Class:

Fields: name (String), unitCost (double), quantity (int).

Parameterized Constructor: Initializes the name, unit cost, and quantity of the cake.

Static Method: calculateTotalCost(Cake[] cakes) – Takes an array of Cake objects, calculates the total cost before discounts, applies the discounts, and returns the final total cost.

Main Class:

Reads input for the number of cakes and details for each cake.

Computes and prints the final total production cost after applying the applicable discounts.

Input format :

The first line contains an integer n, the number of cakes.

For each of the n cakes, the input consists of the following:

A string name (the name of the cake).

A double value unitCost (the cost per unit of the cake).

An integer quantity (the number of units ordered).

Output format :

The output display an double value representing the total production cost after applying the relevant discount rounded to two decimal places.

Refer to the sample output for formatting specifications.

Code constraints :

$1 \leq n \leq 10$

Sample test cases :

Input 1 :

```
4
Cake1
10.0
10
Cake2
5.0
12
Cake3
8.0
8
Cake4
7.0
7
```

Output 1 :

```
218.40
```

Input 2 :

```
2
Cake1
10.0
10
Cake2
5.0
12
```

Output 2 :

```
128.00
```

Input 3 :

```
3
Cake1
```

```
10.0
10
Cake2
5.0
12
Cake3
7.0
7
```

Output 3 :
167.20

Question No: 2

Single File Programming Question
Problem Statement

Samantha, a civil engineer, is designing a park with a central decorative rhombus. To finalize the design, she needs to calculate the area, perimeter, and diagonal product of the rhombus, which will help determine the material required for decoration and fencing, as well as the symmetry of the design.

She creates a RhombusArea class that models the rhombus mathematically, encapsulating properties like diagonal1, diagonal2, area, perimeter, and diagonalProduct.

Formula:

Area = $(d1 \times d2) / 2$
Perimeter = $4 \times \text{side}$, Where $\text{side} = \sqrt{(d1/2)^2 + (d2/2)^2}$
Product = $d1 \times d2$

Note: Use Math.sqrt method to calculate the square root value and Math.pow method to calculate the power value of a number.

Input format :

The first line consists of a double value d1 representing the length of the first diagonal.

The second line consists of a double value d2 representing the length of the second diagonal.

Output format :

The output displays three double-point values, each rounded to two decimal places, in the following format:

Area: <<The area value of the rhombus>>

Perimeter: <<The perimeter value of the rhombus>>

Product: <<The product of the diagonals>>

Refer to the sample output for the formatting specifications.

Code constraints :

$1.0 \leq d1, d2 \leq 10.0$

Sample test cases :

Input 1 :

1.4

2.3

Output 1 :

Area: 1.61

Perimeter: 5.39

Product: 3.22

Input 2 :

4.5
7.6
Output 2 :
Area: 17.10
Perimeter: 17.66
Product: 34.20

PAT 4:-

Question No: 1

Single File Programming Question
Problem Statement

Maya is developing a public transportation fare calculator for her city's transport app. The app needs to calculate the fare for different types of public transport based on the distance traveled and apply discounts for longer distances

City Bus: Charges \$1.50 per kilometer. If the distance exceeds 10 kilometers, a 10% discount is applied.

Urban Metro: Charges \$2.00 per kilometer. If the distance exceeds 15 kilometers, a 5% discount is applied.

Light Rail: Charges \$1.75 per kilometer. If the distance exceeds 20 kilometers, a 7% discount is applied.

Implement a class `PublicTransport` with a method `calculateFare()` and subclasses `CityBus`, `UrbanMetro`, and `LightRail`, each overriding `calculateFare()` to apply the specific fare calculation and discount logic.

Help Maya complete this task by implementing the inheritance and polymorphism.

Input format :

The first line of input consists of a double value `d`, representing the distance traveled.

The second line of input consists of an integer `t`, representing the type of public transport (1 for Bus, 2 for Metro, 3 for Tram).

Output format :

The output should be formatted to two decimal places "Fare: [fare]", where [fare] is the calculated fare based on the distance and transport type if the transport type is valid.

If the transport type is invalid, output: "Invalid transport type".

Refer to the sample output for the formatting specifications

Code constraints :

$1.0 \leq \text{distance} \leq 50.0$

$1 \leq \text{transport type} \leq 3$

Sample test cases :

Input 1 :

5.0

1

Output 1 :

Fare: 7.50

Input 2 :

12.0

1

Output 2 :

Fare: 16.20

Input 3 :

8.0

2

Output 3 :

Fare: 16.00

Input 4 :

18.0

2

Output 4 :

Fare: 34.20

Input 5 :

25.0

3

Output 5 :

Fare: 40.69

Input 6 :

12.7

4

Output 6 :

Invalid transport type

Question No: 2

Single File Programming Question

Problem Statement

You need to develop a system simulating different cryptocurrency wallets and calculating transaction fees and total amounts after transactions. Implement an abstract CryptoWallet class with methods to compute transaction fees, total amount after fees, and yearly fees.

Subclasses include:

BitcoinWallet: 1% (0.01) transaction fee

EthereumWallet: 1.5% t(0.015) transaction fee

AltcoinWallet: 2% (0.02) transaction fee, with additional methods for discounted fees(10%).

Formulas used:

Transaction Fee = Amount × Fee Percentage

Total Amount After Fee = Amount - Transaction Fee

Yearly Transaction Fees = Transaction Fee × 12

Total Amount After Discounted Fee (AltcoinWallet) = Amount - (Transaction Fee × (1 - Discount Percentage))

Input format :

The first line contains an integer walletType representing the type of wallet:

1 for Bitcoin Wallet.

2 for Ethereum Wallet.

3 for Altcoin Wallet.

The second line contains a double amount, representing the amount of

cryptocurrency in dollars.

Output format :

For valid wallet types (walletType = 1, 2, or 3):

The first line contains the transaction fee: Fee: X.XX.

The second line contains the total amount after the fee: Total after fee: X.XX.

The third line contains the yearly transaction fees: Yearly fees: X.XX.

For Altcoin Wallet (walletType = 3), additionally:

The fourth line contains the total amount after applying a discount of 10% to the fee: Total after discount: X.XX.

If the input wallet type is invalid (i.e., not 1, 2, or 3), print: Invalid wallet type

Note: All results are formatted to two decimal places.

Refer to the sample output for the formatting specifications.

Code constraints :

$1.0 \leq \text{amount} \leq 10000.0$

Wallet type Method Choice: 1, 2, or 3.

Sample test cases :

Input 1 :

1

1000.00

Output 1 :

Fee: 10.00

Total after fee: 990.00

Yearly fees: 120.00

Input 2 :

2

500.00

Output 2 :

Fee: 7.50

Total after fee: 492.50

Yearly fees: 90.00

Input 3 :

3

1500.00

Output 3 :

Fee: 30.00

Total after fee: 1470.00

Yearly fees: 360.00

Total after discount: 1473.00

Input 4 :

5

20.0

Output 4 :

Invalid wallet type

PAT 5:-

Question No: 1

Single File Programming Question

Problem Statement

In a messaging application, there is a need to compress repeating characters in user texts to optimize storage and transmission. The application should take a string input from the user and generate a compressed version where each sequence of consecutive identical characters is replaced by the character followed by its count.

Your task is to develop a program that uses `java.util.Scanner` for input handling and `java.lang.StringBuilder` for efficient string creation.

Input format :

The input consists of a string, that includes lowercase letters.

Output format :

The output prints the compressed version of the input string.

Refer to the sample output for the formatting specifications.

Code constraints :

The maximum length of the input string is 100. The input string contains only lowercase characters without spaces.

Sample test cases :

Input 1 :

aabcccccaaa

Output 1 :

a2b1c5a3

Input 2 :

abc

Output 2 :

a1b1c1

Input 3 :

aaabbc

Output 3 :

a3b2c1

Question No: 2

Single File Programming Question

Problem Statement

Jalal is organizing a cooking class and developing an enrollment system to manage sign-ups. Users must input the class size and number of payment instalments with the following rules:

`InvalidClassSizeException`: The class size must be greater than zero. If not, an `InvalidClassSizeException` will be thrown with the message "Class size must be greater than zero".

`MaxParticipantsExceededException`: The maximum number of allowed participants is 30. Exceeding this will trigger a `MaxParticipantsExceededException` with the message "Class size exceeds the maximum."

`PaymentPlanExceededException`: The maximum number of instalments allowed is 6. If the user inputs a number outside this range, a `PaymentPlanExceededException` will be thrown with the message "Payment plan exceeds the maximum".

`MaxFeeExceededException`: The base fee is Rs. 50 per participant and the total fee is calculated as $\text{Total Enrollment Fee} = \text{Base Fee per Participant} * \text{Class Size}$. If the fee exceeds Rs.1500, a `MaxFeeExceededException` will be thrown with the message "Total enrollment fee exceeds the maximum".

After valid input, the system will display the total enrolment fee and instalment amount. The instalment amount is calculated as $\text{total enrolment fee} /$

no. of instalments.

Input format :

The first line of input contains an integer n, representing the class size (number of participants).

The second line contains an integer p, representing the number of instalments for the payment plan.

Output format :

For valid inputs, print the total enrollment fee and the instalment amount for the specified number of months, rounded off to two decimal places.

If there are invalid inputs, appropriate error messages are displayed as follows:

If the class size is not greater than zero: "Class size must be greater than zero".

If the class size exceeds the maximum allowed: "Class size exceeds the maximum".

If the number of installments is less than 1 or exceeds the maximum allowed: "Payment plan exceeds the maximum".

If the total enrollment fee exceeds the maximum limit: "Total enrollment fee exceeds the maximum".

Refer to the sample output for the formatting specifications.

Sample test cases :

Input 1 :

10

5

Output 1 :

Total enrollment fee: 500.00

Installment amount for 5 months: 100.00

Input 2 :

35

Output 2 :

Class size exceeds the maximum

Input 3 :

-5

Output 3 :

Class size must be greater than zero

Input 4 :

20

8

Output 4 :

Payment plan exceeds the maximum

Input 5 :

29

6

Output 5 :

Total enrollment fee: 1450.00

Installment amount for 6 months: 241.67

Input 6 :

12

1

Output 6 :

Total enrollment fee: 600.00

Installment amount for 1 months: 600.00